

A finite noise window for concentration of a shared flow

Deterministic disturbance and steps toward selection

David Galbraith

Revised draft, 9 September 2026

Abstract

Can variation, unequal flow sharing and reproduction be connected within a physical description of selection? We examine a sequence of models in which unresolved deterministic dynamics supply effective disturbance, a bounded range of disturbance scaling permits concentration, and flow can fund copying. State-dependent diffusion changes how often a channel occupies a given state. When channels share a fixed total throughput, their population distribution and their flow distribution can differ substantially. We study a bounded, reflected Itô model with independent noise, share proportional to g_i^m , and diffusivity $D_i = N g_i^q / \sum_j g_j^m$. Its exact joint stationary density is proportional to $(\sum_j g_j^m) \prod_i g_i^{-q}$. The tagged-channel density is $p_N = (1 - 1/N)p_0 + f/N$, where $p_0 \propto g^{-q}$ and $f \propto g^{m-q}$; f is also the pooled flow density. The population fraction above the pooled half-flow threshold tends to zero as both state range and population grow if and only if $1 \leq q \leq m+1$, for exact powers. At fixed population it instead approaches $1/(2N)$ inside the window. The upper boundary is the familiar weighted-moment threshold, here given an exact conserved-flow interpretation. It limits a specific noise-driven concentration mechanism; it is neither a universal noise limit nor a result about persistent winners. Independent event simulations check the stationary predictions on bounded grids. Separate lattice and copying experiments illustrate further steps in the proposed connection. An inherited-state branching extension predicts a copying-cost threshold and, under activity-dependent carrier damage, a reversal of restoration's effect as transit grows. These conditional predictions use established mathematics. The models do not yet derive channel emergence, the full window and reproduction in one physical system. Empirical applicability remains untested.

1 The physical question and the upper boundary

The motivating proposal is conditional: disturbance associated with a driven flow could help produce concentrated sharing only within a suitable scaling window, and that concentration could become relevant to selection if states persist and are copied. The window is essential to this proposal. A mechanism that concentrates under every disturbance rule would not establish it.

The research programme asks whether this sequence can be constructed without adding random forcing as a fundamental ingredient: deterministic fine-scale dynamics produce effective variation; variation affects the division of flow; and inherited states use some of that flow to reproduce. This paper separates an exact stationary result from computational thought experiments that explore those connections. The proposed synthesis is not equivalent to the known power-law calculation on which one step depends.

A collection of paths can be counted in two ways: by the number of paths in each state, or by the fraction of throughput they carry. A rare large state can matter little to the first count and greatly to the second. The question here is when that difference becomes extreme as the accessible state range grows.

Noise affects this comparison through residence times. In a scalar driftless Itô diffusion with reflecting boundaries, the stationary density is inversely proportional to diffusivity. State-dependent noise and occupation probabilities have a long history [1, 2]. Weighting a power-law distribution by a power of its argument introduces a standard moment threshold [3]. Neither ingredient is new.

Newman explicitly discusses both the general moment condition and the fraction of a weighted total held in a population tail (Secs. III.B and III.D of Ref. [3]). Their use for a fixed shared flow requires care: each channel's share depends on every other channel, and the same normalization can change the physical clock of the state dynamics. We give an exactly solvable coupled model in which this dependence is explicit. The result identifies the finite-population correction and makes the observable behind the noise window precise.

For $m = 1$, the upper edge is $q = 2$. Beyond it, increasing the upper state cutoff does not make a vanishing population fraction account for half the pooled flow. This does not mean that flow becomes equally distributed. A small but nonzero fraction can still carry half. Nor is q a noise amplitude: it describes how diffusivity changes with state. Multiplying every diffusivity by one constant changes relaxation times, but leaves all stationary results below unchanged.

2 A model with conservation imposed at every instant

Let $N \geq 1$ channels have dimensionless states $g_i \in [1, R]$, where $R > 1$. The model specifies possible flow routes and their response law; it does not prescribe which routes become dominant. The lattice experiment in Sec. 6 tests differentiation from identical coarse-route states. Spatial formation and connectivity of routes in a material are abstracted out: no spatial energy-gradient field or changing network topology is modeled here. For $m > 0$, define the total weight S and the actual flow through channel i by

$$S(\mathbf{g}) = \sum_{j=1}^N g_j^m, \quad J_i(\mathbf{g}) = I \frac{g_i^m}{S(\mathbf{g})}. \quad (1)$$

Thus $\sum_i J_i = I$ for every configuration, not only in expectation. We choose the interior dynamics

$$dg_i = \sqrt{2D_i(\mathbf{g})} dW_i, \quad D_i(\mathbf{g}) = \frac{N g_i^q}{S(\mathbf{g})}, \quad (2)$$

with independent Wiener processes, normal reflection at both endpoints, and zero interior drift. Boundary local-time terms are implicit. A positive common prefactor has been absorbed into the time unit. The lower cutoff sets the state unit; all powers refer to this specified coordinate.

The normalization in Eq. (2) is a constitutive assumption. On the line $q = m$, $D_i = (N/I)J_i$, so noise variance rate is proportional to actual flow. Away from that line the model keeps the common factor $1/S$. The apparently similar choice $D_i \propto J_i^{q/m}$ instead contains $S^{-q/m}$ and defines different dynamics. Conservation of flow alone does not select either rule.

The Itô convention would follow from an appropriate small-kick limit of conditionally unbiased additive events in g , evaluated at the pre-event state. Such a limit also needs control of temporal correlations and higher moments. Unresolved deterministic dynamics do not automatically satisfy these conditions. Under a nonlinear change $y = h(g)$, Itô's formula adds the drift $D(g)h''(g)$ even when the drift in g is zero. The zero-drift assumption must therefore be justified in the measured coordinate, rather than restored after arbitrary relabelling.

3 Exact stationary population and flow distributions

On $[1, R]$, write

$$Z_R = \int_1^R g^{-q} dg, \quad p_0(g) = \frac{g^{-q}}{Z_R}, \quad \mu_R = \int_1^R g^m p_0(g) dg, \quad f_R(g) = \frac{g^m p_0(g)}{\mu_R}. \quad (3)$$

All quantities are finite at finite R , for every real q .

Proposition 1. *The unique stationary density of Eq. (2) on the reflecting box is*

$$\pi_{N,R}(\mathbf{g}) = \frac{S(\mathbf{g})}{N\mu_R} \prod_{j=1}^N p_0(g_j). \quad (4)$$

Its tagged-channel marginal and its pooled flow density are, respectively,

$$p_{N,R}(g) = \left(1 - \frac{1}{N}\right) p_0(g) + \frac{1}{N} f_R(g), \quad (5)$$

$$\mathbb{E}_\pi \left[\sum_{i=1}^N \frac{J_i}{I} \delta(g - g_i) \right] = f_R(g). \quad (6)$$

Proof. Integration of $S \prod_j p_0(g_j)$ gives $N\mu_R$, establishing normalization. For each coordinate,

$$D_i \pi_{N,R} = \frac{1}{Z_R \mu_R} \prod_{j \neq i} p_0(g_j)$$

is independent of g_i . Every probability current $-\partial_i(D_i \pi_{N,R})$ therefore vanishes, including at the boundaries. The diffusion is uniformly nondegenerate on the connected compact box, giving uniqueness. Expanding the sum S and integrating out the other coordinates gives Eq. (5). Multiplication by $J_i/I = g_i^m/S$ cancels S in the joint law; integration and summation give Eq. (6). $\square$

The calculation also has a simple clock interpretation. If $L_0 = \sum_i g_i^q \partial_i^2$ is the generator of independent reflected channels, the coupled generator is $L = (N/S)L_0$. Configurations with larger S are traversed more slowly relative to that independent process; their stationary weight acquires the factor S . This explains both the solvability and the dependence of the result on the chosen normalization.

An exact independent sampling construction is useful: choose one channel uniformly, draw its state from f_R , and draw the other $N - 1$ states independently from p_0 . Averaging over the chosen channel reproduces Eq. (4). That designation is resampled for each configuration; it does not identify a permanent winner.

The correction matters even in simple cases. For $N = 1$ and $q = m$, the channel carries all the flow and has constant diffusivity. Its stationary density is uniform, not g^{-m} . At fixed R , the tagged marginal approaches p_0 as $N \rightarrow \infty$. The independent-channel calculation is recovered in that limit.

4 The concentration observable and its window

Let a_R be the state threshold above which half of the pooled flow lies:

$$\int_{a_R}^R f_R(g) dg = \frac{1}{2}. \quad (7)$$

Define $P_0(R) = \int_{a_R}^R p_0(g)dg$ and $P_N(R) = \int_{a_R}^R p_{N,R}(g)dg$. Equation (5) gives

$$\boxed{P_N(R) = \left(1 - \frac{1}{N}\right) P_0(R) + \frac{1}{2N}.} \quad (8)$$

This counts a population fraction above one threshold fixed across stationary observations. It is a pooled observable, with each observation weighted equally in physical time.

Theorem 1. *For Eqs. (1)–(2) with exact powers and $m > 0$, $P_N(R) \rightarrow 0$ as both $R \rightarrow \infty$ and $N \rightarrow \infty$ if and only if*

$$\boxed{1 \leq q \leq m + 1.} \quad (9)$$

At fixed N , the limit inside this window is $1/(2N)$.

Proof. Put $A = m - q + 1$ and $B = 1 - q$. Direct integration gives

$$a_R = \begin{cases} [(R^A + 1)/2]^{1/A}, & A \neq 0, \\ \sqrt{R}, & A = 0, \end{cases} \quad P_0(R) = \begin{cases} (R^B - a_R^B)/(R^B - 1), & B \neq 0, \\ \log(R/a_R)/\log R, & B = 0. \end{cases} \quad (10)$$

For $q < 1$, $a_R/R \rightarrow 2^{-1/A}$ and $P_0 \rightarrow 1 - 2^{-B/A} > 0$. At $q = 1$, $P_0 \sim \log 2/(m \log R) \rightarrow 0$. For $1 < q < m + 1$, a_R grows proportionally to R , while the normalized population tail above it vanishes. At $q = m + 1$, $P_0 = 1/(R^{m/2} + 1) \rightarrow 0$. For $q > m + 1$, $a_R \rightarrow 2^{1/(q-m-1)}$, so $P_0 \rightarrow 2^{-(q-1)/(q-m-1)} > 0$. Equation (8) supplies the finite- N result and the joint limit. $\square$

Either iterated limit gives this classification, as does any sequence with both R and N tending to infinity. Stationarity is taken at each bounded pair (N, R) first. No unbounded diffusion or uniform equilibration time is asserted. The same classification holds for any fixed pooled flow fraction strictly between zero and one, with the corresponding fraction divided by N replacing $1/(2N)$.

The integrals behind the two edges are $\int g^{-q}dg$ and $\int g^{m-q}dg$. At the lower edge the population normalization is logarithmic; at the upper edge the flow normalization is logarithmic. The upper-edge population fraction itself decays as a power of R . Figure 1 displays the finite-range expressions.

4.1 Pooled concentration is not a snapshot winner count

In a single configuration, rank shares in descending order and count the smallest number of whole channels whose shares sum to at least one half. Call that count divided by N H_N . It always satisfies $H_N \geq 1/N$. In contrast, P_N can approach $1/(2N)$ because its fixed threshold need not carry half the flow in every snapshot. These observables must not be substituted for one another. The numerical checks measure $\mathbb{E}H_N$ separately against the exact joint sampler. Neither statistic measures how long a particular channel remains dominant. Stationary concentration alone establishes neither persistent structure nor Darwinian selection.

For example, at $m = 1$, $q = 2$, $R = 10^8$, $P_0 \simeq 0.000100$, whereas $P_{40} \simeq 0.012598$. This is a substantial relative correction to the tail fraction, even though $1/N$ is small.

4.2 Limiting slopes do not determine the endpoints

The inclusive endpoints require exact powers or additional tail assumptions. On $g \geq e$, consider the unnormalized flow weight $w(g) = g$ and noise factor $d(g) = g^2(\log g)^2$. Their limiting log-log slopes are $m = 1$ and $q = 2$, but $\int_e^\infty w(g)/d(g)dg$ converges. The upper-edge separation fails. This example also fits the coupled construction with $S = \sum_i w(g_i)$ and $D_i = Nd(g_i)/S$: replace p_0 by the normalized $1/d$ and f_R by the normalized w/d in the proof. Logarithmic corrections do not change the strict interior power inequalities, but can decide the boundary cases. An estimated slope near an edge is consequently insufficient to classify a measured system.

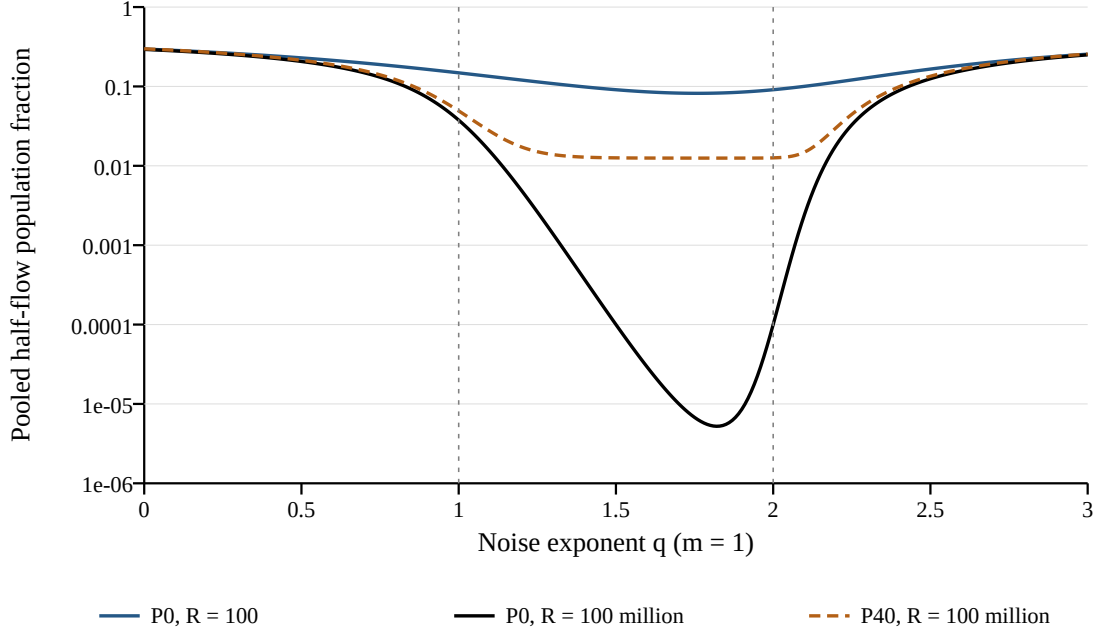


Figure 1: Exact stationary predictions, not empirical measurements or large-range dynamic simulations. The asymptotic window is between the dotted lines. Finite range smooths both boundaries; finite population supplies a floor. P_0 is the large-population limit at each fixed R . The curve at $R = 10^8$ is not the infinite-range limit.

5 What would make the result physically useful?

The upper bound excludes a proposed mechanism when the measured disturbance scaling is too steep relative to the flow weight, provided the rest of the model applies. It also distinguishes state-dependent disturbance from monotonic reinforcement intuition. At $m = 1$, the large-range, large-population half-flow fraction vanishes for $q = 2$, but approaches $1/8$ for $q = 2.5$. Above the boundary it can still be very small, especially close to $q = 2$. The mathematical transition is the loss of a vanishing fraction, not an abrupt return to equal sharing in a finite apparatus.

The basic inequality is a standard power-law moment condition [3]. Its value here is the connection to actual conserved shares and the observable correction in Eq. (8). These make a conditional prediction that can be checked. They do not establish a new general law of nature, and no priority claim for the elementary threshold is made.

One possible microscopic route to state-dependent variance uses contact patches. If each patch contributes an unbiased additive perturbation in the same measured coordinate, n independent patches give variance proportional to n ; perfectly coherent patches within a channel give variance proportional to n^2 . Inferring $q = m$ or $q = 2m$ additionally requires $n \propto g^m$, fixed coupling per patch, and a suitable time scale. It does not derive the common $1/S$ clock in Eq. (2). In particular, a rule proportional to squared actual flow would have a different normalization. Coherence within a channel must also be distinguished from correlations between channels, which violate the independent-noise assumption.

A fair physical test needs simultaneous channel states and individual flows, with a fixed total and a stable channel population. It must estimate both conditional drift and variance rate in the chosen coordinate, check cross-channel correlations, and allow enough relaxation before comparing stationary statistics. For Eq. (2), conditioning on the configuration or accounting for S is necessary: a regression on g_i alone need not recover q . Drift, measurement noise, jumps,

changing populations and limited state range can each prevent the test from addressing this model. Power-law fits themselves also need checks against alternatives [4].

A useful experiment would vary the state dependence of disturbance across the upper edge while measuring the finite-range prediction, rather than expecting an infinite-limit discontinuity. No physical dataset is used as validation in this paper.

6 Thought experiments toward selection

6.1 Effective noise in a deterministic world

The first construction, `sims/core/08`, puts each of forty coarse routes over a block of a coupled logistic-map lattice. The lattice evolves deterministically. A route responds to the difference between its block mean and a slowly updated running mean, with a coupling proportional to its share. Random numbers initialize the lattice and routes; none are drawn during their subsequent evolution. Fixing the complete initial state fixes the trajectory.

In the retained default run, the chaotic lattice gives a final top-three flow share of 0.62, compared with 0.08 for a frozen lattice. Its sampled block means have mean lag-one autocorrelation about 0.01, and the repeated run reports identical summary statistics. These are demonstrations of deterministic driving with irregular coarse fluctuations and concentrated flow, not a proof that the coarse process is Markovian or converges to the Itô model above.

An equal-start control removes the initial differences between coarse routes by setting every $g_i = 1$, while retaining microscopic variation in the fine lattice. After 60,000 deterministic updates, the top-three shares are 0.615, 0.463 and 0.529 for three fine-state seeds, from $3/40 = 0.075$ initially. The matched frozen controls remain at 0.075. Fine-scale dynamics therefore seed differentiated sharing without initial coarse-channel size differences in these runs. The possible routes and coupling rule are specified, but the dominant routes are not supplied as an initial pattern. This is a test of seeding within that geometry, not a test of the full two-sided window.

The periodic control is informative: it gives an even larger top-three share, 0.95. The experiment therefore does not show that chaos is necessary, despite an earlier description of that control. Low autocorrelation alone would not establish unpredictability from all coarse information either. The contribution of the lattice experiment is to make the absence of ongoing random input explicit and reproducible within a flow-sharing example.

Deterministic dynamics admitting stochastic effective descriptions are already established mathematically [7, 8]. Their role here is as a candidate physical source for the disturbance, not a new discovery that determinism and statistical noise can coexist. A diffusion limit for this particular lattice, including its effective drift and stochastic convention, remains to be derived. This matters because deterministic multiplicative forcing need not produce a driftless Itô limit [8].

6.2 Contact geometry and the window are separate tests

The next construction, `sims/core/09`, recruits more lattice patches for a route with a larger share, keeping coupling per patch fixed. In the retained run, the variance-versus-mean-share slope is 1.01 for sums over separate patches. A common coherent patch gives slope 2.00; using fixed patches with amplitude proportional to share gives 1.99. These are across-route diagnostic slopes, not direct estimates of conditional infinitesimal diffusivity.

The final top-three shares are 0.17, 0.08 and 0.63, respectively. The coherent case uses the same fluctuation for all routes and so changes cross-route correlation as well as variance scaling. It cannot isolate the effect of moving q to the upper edge. Both slopes near two are also at the $m = 1$ boundary, rather than beyond it. The lattice examples consequently support a

mechanism for effective disturbance and its scaling; they do not demonstrate the full two-sided window in a single deterministic substrate.

The exact theorem answers a different, controlled question: for the specified diffusion, when does pooled concentration vanish as a population fraction? The lattice routes have a lower clipping rule and no fixed upper reflecting cutoff, and their effective drift and shared clock have not been matched to Eq. (2). A next test must connect those models before treating their results as one experiment.

6.3 From unequal sharing to differential reproduction

Concentration alone contains no lineage. Suppose each route also has a store of usable work A_i , increasing at rate $\dot{A}_i = \eta J_i$, and a copy costs E_{copy} . With constant flow and no intervening loss, the time between copies is $E_{\text{copy}}/(\eta J_i)$. For time-varying flow, accumulated work sets the number of completed copies, up to the remainder in the store. Thus the ideal long-run rate is $b_i = \eta \bar{J}_i / E_{\text{copy}}$. This derives the rate conditional on a copying mechanism, an efficiency and a cost; it does not derive the existence of copying machinery.

In `sims/core/18`, a funded copy replaces a randomly chosen route with an inherited state plus a perturbation. The default run with copy cost 200 ends at mean state 8.67 ± 1.27 , compared with 4.32 ± 1.17 at cost 500 and 1.05 ± 0.19 without copying. These are means and population standard deviations across three runs, not confidence intervals. Fitted copying-rate slopes are 0.0045 and 0.0017, against ideal values 0.0050 and 0.0020; replacement can erase stored resources, and self-targeted attempts are counted.

Here the connection to selection becomes explicit: state varies, is inherited, and affects reproduction through throughput. The simulation supplies inheritance and replacement rules and uses random draws during evolution. It is therefore a separate thought experiment, not a completed demonstration of selection driven entirely by the deterministic lattice. Other repository models investigate whether copying refreshes memory faster than restoring dynamics erase it. Those address another necessary condition for the proposed sequence, rather than following automatically from the concentration theorem.

6.4 The existing heredity sequence

The original repository distinguishes renewal of a pattern, reproduction across lineages, and transfer through a separate carrier. A whirlpool illustrates continuity of organisation through material turnover. Heredity of a variant asks whether a particular deviation survives that renewal. Boundary models explore maintained loops and reconstruction from a persistent bed; they do not establish a general law of heredity. Channel-junction correlations also reflect spatial continuity. A confluence viewed backwards is not itself an observed reproductive event. Turbulent puffs that split and decay provide a physical comparison for reproduction versus loss [13], without establishing inheritance of selectable variants in the present model.

Core 12 supplies copying and tests preservation against restoration. Core 13 already implements detached tokens affecting another site's response; core 15 examines transit. These are existing toy constructions, not absent mechanisms. Their transmission and response laws are assumptions. A small catalytic configuration could carry a property without carrying an entire channel, and could itself perform the catalytic function. Schrödinger's aperiodic crystal is a historical image for stored arrangement [9], not a copying mechanism. Autocatalytic RNA experiments distinguish reproduction from persistence of composition [10]; synthetic self-replicators selected for photocatalytic function provide a chemical comparison [11].

Deterministic disturbances become mutations when they alter a transmissible state that reaches descendants. Temporary channel fluctuations need not do so. The new checks below use explicit random draws; the deterministic lattice and inherited carrier have not yet been joined in one construction.

6.5 Establishment with inherited states

The old lineage instrument counted births at a site, allowing confusion between the founder and later occupants. It now tracks individual identity. Survival at a finite horizon cannot establish eventual survival. Nor does a failed scalar founder-birth count invalidate branching theory.

The new calculation uses a finite trait lattice with row transition generator T , birth-attempt rates b_i , disappearance rates $d_i > 0$, and a substochastic offspring matrix P . Here P_{ij} is the probability an attempt by type i produces a viable daughter of type j ; missing row mass means a failed copy. Parents survive birth events. In a fixed background the probability $Q_i(t)$ of extinction by time t , from one individual, obeys

$$\dot{Q} = TQ + b \odot Q \odot (PQ - P\mathbf{1}) + d \odot (\mathbf{1} - Q), \quad Q(0) = 0, \quad (11)$$

where $\odot$ means elementwise multiplication. With perfect inheritance $P = I$, the birth term is $b_i(Q_i^2 - Q_i)$. Eventual extinction is the minimal fixed point. This is established multitype branching mathematics [12]. For the finite irreducible, nonsingular process used here, positive eventual survival is equivalent to

$$\rho(K) > 1, \quad K = (\text{diag } d - T)^{-1} \text{diag}(b)P. \quad (12)$$

With $b_i = \eta J_i / E_{\text{copy}}$, the corresponding copying-cost threshold is

$$E_{\text{copy}} < E_{\text{critical}} = \eta \rho[(\text{diag } d - T)^{-1} \text{diag}(J)P]. \quad (13)$$

This applies known mathematics to measured flow, state change, loss and transmission under the stated approximation. Exponential birth clocks are assumed; a deterministic accumulator needs an additional work or age state. The background is fixed while a lineage is rare, so this is not a prediction of takeover of a finite shared flow.

Analytic one-state birth/death cases check the implementation. Another 18,000 exact-event trials check finite-time extinction, and 12,000 trials examine both sides of the threshold. Cap hits are censored rather than counted as survival. All 30 predictions lie within marginal 99% Wilson intervals widened for censoring. These are not simultaneous bounds or proof of continuum convergence. Raw outcomes, timestep refinement and hashes are in `docs/heredity_results/`.

6.6 A conditional reversal of the threshold

The retained extension takes $g \in [1, 3]$, diffusion $.02g$, restoration $-k(g - 1)$, disappearance $.4$, and birth attempts cg . It assumes viable transmission probability $\exp(-.02g\tau)$ over transit time τ : greater activity also damages the carrier, with a damaging event preventing a viable copy. This is an additional hypothesis, not a consequence of the noise exponent. Transit survival is applied at birth without an explicit travel delay.

On 64 grid intervals, restoration $k = .7$ raises the critical copying coefficient from $.1747$ to $.3517$ at $\tau = 0$, but lowers it from 1.5880 to 1.3803 at $\tau = 60$. The sign reversal also occurs on 16 and 32 intervals. Replacing the damage rate $.02g$ by the constant $.02$ removes the reversal at every tested transit time (Fig. 2). Restoration sacrifices flow but can improve transmission enough to help establishment. For a fixed trait the viable birth rate $cg \exp(-.02g\tau)$ peaks at $g = 1/(\tau)$ if that lies in the allowed range. The branching calculation includes changes of trait during life.

This is a candidate prediction of the specified couplings, distinct from the stationary concentration window. Its originality is not established. Reproduction-fidelity trade-offs already have experimental precedents [14]. A physical test must measure carrier damage rather than infer it from channel diffusivity alone.

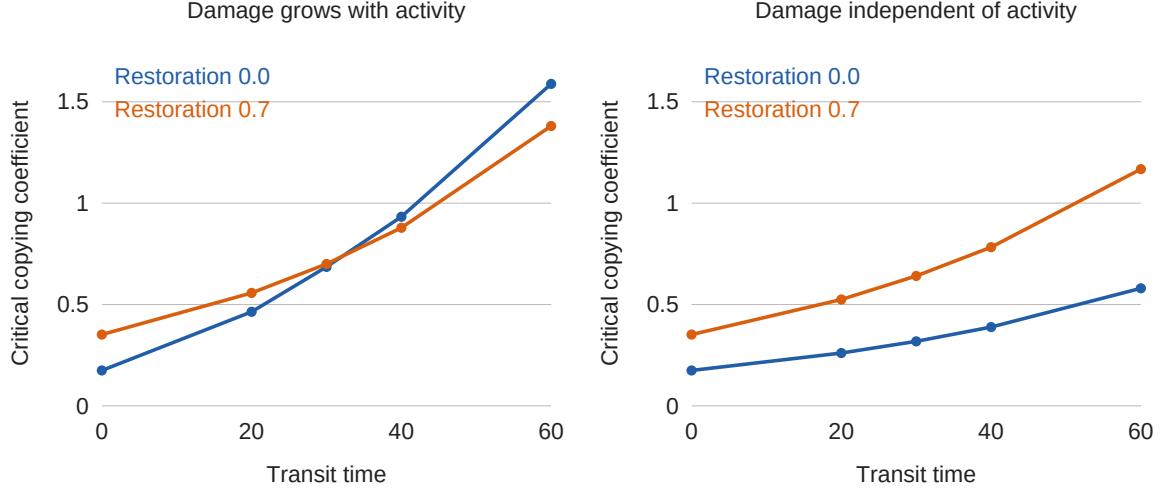


Figure 2: Critical copying coefficient on 64 trait intervals. Lower values make establishment easier. Points are numerical calculations; lines guide the eye. Activity-dependent damage allows restoration to change from a disadvantage to an advantage as transit grows. The constant-damage control does not. These are model predictions, not empirical observations.

6.7 Matched transfer and repeated reproduction

The new assay gives binary and continuous carriers identical initial values $h = \pm 1$ and the same destination response $g = 1 + .5 \tanh(h)$. Destinations are fresh: no original channel state or work store accompanies the carrier. Binary carriers flip at rate .005; continuous carriers follow an Ornstein–Uhlenbeck law with noise amplitude .1 and restoration .005 or .3. Exact transition sampling avoids timestep error. At transit time 40 the destination response difference between source types averages .510 for binary, .578 for slow continuous, and approximately zero for fast continuous carriers (ten seeds of 10,000 packets). Permutation effects are near zero. Continuous storage can transmit state over these finite times; universal gating necessity is not supported. Physical storage costs are not matched.

A separate generational model funds 256 unit-cost births from total flow one over 256 time units. Parent opportunities are proportional to flow, or equal in a neutral control; only carrier state is transmitted and each site’s response is rebuilt. After 40 generations, positive-state fractions average .919 for binary and .966 for slow continuous carriers, versus .483 and .486 in the neutral control (20 seeds). Fast continuous carriers remain near .5. This assumes a common funding pool and copying rules. Shuffling the recorded parent–child pairing preserves population trajectories while removing their association. It is a statistical null: a rising mean alone cannot identify inheritance. Carrier origins and physical copying remain open.

6.8 What the combined proposal commits to

The scientific proposal is the connection between these steps, including the bounded concentration condition. The theorem constrains this concentration mechanism, not every possible route to Darwinian selection. It would gain substance from a construction in which the same physical rules supply effective variation, cross both noise boundaries, preserve inherited differences and fund reproduction. The equal-start lattice control demonstrates a seed mechanism for differentiation among initially equivalent possible routes. Extending that mechanism to a material whose spatial flow paths and connectivity also develop requires a spatial model of that response. What remains incomplete is the joining of seeding, the controlled window test, memory and reproduction in one construction. Nor does the prior art cited for the separate

N	Grid intervals	Occupancy CDF	Flow CDF	Mean snapshot fraction
8	4	0.017014	0.021730	0.011423
8	8	0.020299	0.024502	0.005979
40	8	0.009496	0.008539	0.001585

Table 1: Largest final absolute errors across the six exponent pairs and four initial conditions for each retained run, all on $[1, 3]$. CDF errors compare with the exact lattice law. The snapshot reference uses 100,000 independent joint-law samples per pair. Seeds are pooled at the final observation time.

ingredients establish that the complete proposed mechanism is already known.

The default outputs and equal-start controls are retained with source hashes and reproduction commands in `docs/research_examples/`. They report finite-time toy-model behaviour. They neither establish empirical validity nor replace the stationary checks below.

7 Numerical checks of the stationary result

The repository implements a continuous-time nearest-neighbour chain on a uniform state lattice with spacing h . Each allowed jump of channel i by $+h$ or $-h$ has rate

$$\lambda_i(\mathbf{g}) = \frac{Ng_i^q}{S(\mathbf{g})h^2}. \quad (14)$$

Outward boundary jumps are absent. Exponential holding times and rate-weighted event selection simulate this finite chain without a time-step approximation. The lattice invariant law is exactly Eq. (4) with sums replacing integrals. Grid refinement remains necessary for a continuous-state comparison. Observations are taken at fixed physical times, because sampling only jump events would bias residence-time statistics.

Eight automated tests include detailed balance on every edge of an enumerated three-channel, three-state system; independent marginal calculations; the single-channel case; the finite-population quantile identity; sampling checks; continuum integral comparisons; grid refinement; and relaxation from both boundaries in a simple control. The retained ensemble runs use six exponent pairs: $(1, 0.5)$, $(1, 1)$, $(1, 1.5)$, $(1, 2)$, $(1, 2.5)$ and $(2, 4)$. For each pair they use lower-boundary, upper-boundary, log-uniform and exact stationary initial conditions, with three seeds and 128 independent ensembles per seed. Physical observation times are 0, 1, 4, 16.

All retained final errors are below the declared tolerance of 0.05 (Table 1). This is an acceptance criterion, not a confidence interval. Separate exact-law calculations check grid refinement and evaluate large state ranges; they are not dynamical runs at those ranges. The checks support the implementation on the tested bounded grids. They do not establish large-range equilibration times or empirical applicability.

The earlier independent-walker check is excluded from this evidence. It started log-uniformly, which is already the target stationary law at $q = 1$, and did not establish relaxation. It is retained in the repository as a labelled legacy diagnostic. The exact power integrals remain valid.

8 Drift and related models

Scalar diffusion with drift has the zero-current law

$$p(g) \propto \frac{1}{D(g)} \exp\left(\int^g \frac{v(x)}{D(x)} dx\right). \quad (15)$$

For $D = d_0 g^q$ and $v = a g^{q-1}$, this shifts the scalar exponent to $q - a/d_0$. It does not give a general drift correction for the coupled model above. Likewise, scalar accumulation near an upper cutoff does not prove that one channel takes over a shared total. Coupled drift and winner persistence remain separate problems.

Preferential attachment [5] concerns growing networks with attachment rules. Its linear boundary is not a derivation of this model’s lower noise edge. Wealth condensation [6] couples exchange and multiplicative growth; the variable being condensed and its dynamics differ from the fixed-throughput observable used here. These are related comparisons, not equivalent mechanisms.

The repository audit distinguishes the lineage and transport checks discussed here from older exploratory models and unreproduced historical summaries. Their copying, storage and loss rules remain assumptions separate from the exact stationary result.

9 Conclusion

For the specified reflected conserved-flow diffusion, the exact stationary law gives a finite noise window and a finite-population correction. The upper bound states when rare large states cease to dominate pooled throughput in the large-range limit. Its algebra is established moment mathematics. The paper makes that interpretation exact for a coupled model and places it within a proposed route from deterministic disturbance to selective reproduction. The lattice and copying examples give concrete steps in that route, while also exposing the missing connections. The inherited-state extension supplies a copying-cost threshold and a candidate reversal of restoration’s effect when carrier damage grows with activity. These conditional predictions use established branching mathematics. The scientific question is whether the steps can be joined under one physically justified set of rules. Established mathematics does not answer that question by itself, and the separate examples do not yet amount to a derivation of natural selection from an energy gradient.

Code and reproducibility. The editable manuscript, compiled PDF, figure data and generation script are in `paper/` at <https://github.com/divadwg/selection-as-physics>. The stationary implementation and tests are in `sims/validation/` and `tests/`; retained data are in `docs/validation_results/`. The underlying validation was committed as `e802c6f`. Build and run instructions are in `paper/README.md`. This revision supersedes the August 2026 draft’s uncorrected finite-population claim, limiting-slope endpoint claim, independent-walker convergence claim and coupled-takeover interpretation.

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